

RUBRIC-BASED EVALUATION AND AGENTIC REFINEMENT FOR INTENT-ALIGNED TEXT-TO-MUSIC GENERATION

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ABSTRACT

Text-to-music systems can generate increasingly plausible and diverse audio, yet determining whether they faithfully realize user intent remains challenging. Existing evaluation methods primarily measure overall quality or coarse-grained text-audio relevance, obscuring alignment failures in specific musical requirements and implicit intent. We formulate text-to-music intent alignment as *prompt-specific musical rubric satisfaction*, representing each request as a set of independently verifiable requirements covering both explicit and completed musical intent. We introduce MuRA-Bench, a fine-grained benchmark for musical intent alignment derived from real-world platform requests and revised by music experts. MuRA-Bench provides interpretable evaluation at the levels of individual requirements, intent sources, and musical dimensions. Building on this formulation, we propose MIRA (*Musical Intent Refinement Agent*). MIRA uses external musical information to complete the task specification, converts item-level judgments from a music verifier into fine-grained environment observations, and adaptively searches over candidate prompt states under a bounded generation budget. Its tree-structured memory records requirement-level improvements and regressions associated with different actions, allowing accumulated feedback to inform subsequent search branches. Expert agreement experiments establish the correlation between our item-level automatic evaluation and human musical judgments. Experiments across open-source and commercial generation backends further show that MIRA enables an open-source generator to match some representative commercial systems in intent alignment.

1 INTRODUCTION

Text-to-music generation has advanced from short caption-conditioned clips to high-fidelity music controlled by natural-language descriptions Agostinelli et al. (2023); Copet et al. (2023); Liu et al. (2024); Melechovsky et al. (2024). Recent systems further support lyrics, reference audio, long-form structure, and language-model planning, bringing increasingly complete songs within reach of open-source models Liu et al. (2025b); Lei et al. (2024); Yang et al. (2025); Gong et al. (2026); Lei et al. (2025; 2026). In creative workflows, plausible audio must also realize user intent. Real-world requests mix explicit constraints on vocals, rhythm, instrumentation, structure, or mood with references to an artist, work, scene, or use case as shorthand for harder-to-verbalize stylistic and production choices. Following these requests requires satisfying both stated constraints and contextually implied intent.

Existing evaluation methods only partially capture this problem. Distributional, cross-modal, and audio-language metrics primarily measure audio quality or broad text-audio relevance Kilgour et al. (2019); Elizalde et al. (2023); Deshmukh et al. (2024); Herremans & Roy (2026). Expert-rated datasets, human-preference platforms, learned quality metrics, and specialized benchmarks provide stronger signals for perceptual quality, preference, and professional musical dimensions Liu et al. (2025a); Kim et al. (2025); Zhu & Li (2026); Ma et al. (2026); Wu et al. (2026). These evaluations support system comparison but typically use holistic scores or shared dimensions, without specifying individual request requirements. A high-scoring track may still omit a required instrument, violate a negative constraint, misinterpret a use case, or miss traits implied by a reference. Recent

audio evaluation methods address part of this limitation by decomposing instructions into verifiable questions or binary rubric items Kuan et al. (2026); Li et al. (2026). However, they focus primarily on general text–audio semantics or instruction following across broad audio domains, rather than expert-revised musical criteria derived from real-world music requests.

We therefore formulate text-to-music intent alignment as *prompt-specific musical rubric satisfaction*. Each request is represented by a set of independently auditable rubric items covering both explicitly stated constraints and musical intent completed from references and context. The specification varies with the request, revealing which requirements are satisfied or missed alongside overall alignment. We instantiate this formulation in **MuRA-Bench**, a public benchmark containing 100 controlled music-generation requests derived from anonymized intent seeds collected on the Mureka platform. For each request, a language model drafts a musical rubric that is subsequently reviewed and revised by six music experts. Named references are converted into generalized, audibly verifiable, and copyright-conscious musical traits rather than retained as direct imitation targets.

The same rubric perspective also provides an actionable signal for improving generation. Existing methods strengthen prompt following through pre-generation planning or model-specific inference-time control, including reward-guided search, latent optimization, and internal intervention Gong et al. (2026); Novack et al. (2024); Koo et al. (2025); Roy et al. (2025). We focus on identifying unmet request-specific requirements in completed tracks and converting them into subsequent repairs. We propose **MIRA**, a *Musical Intent Refinement Agent* that organizes text-to-music generation as verifier-guided search. MIRA uses the request and tool-grounded musical evidence to construct an online rubric, converts generated audio into requirement-level observations with a music-specialized verifier, and explores alternative prompt states under a bounded generation budget. Feedback is retained along the search structure, allowing subsequent actions to preserve satisfied requirements while targeting unresolved ones. Operating outside the generator, MIRA supports structurally different black-box backends without parameter updates.

We validate item-level scores against expert judgments, then compare generators on stated and completed musical intent. We evaluate MIRA on two open-source backends under different generation budgets, analyze its major components, and use commercial systems as alignment references. The results show that request-specific evaluation exposes differences obscured by global relevance scores. MIRA consistently improves intent alignment and enables an open-source generator to match or surpass some representative commercial systems under MuRA-Bench evaluation.

Our work makes the following contributions:

- We formulate text-to-music intent alignment as *prompt-specific musical rubric satisfaction*, extending evaluation from global relevance to item-level verification of concrete, request-specific musical requirements.
- We introduce **MuRA-Bench**, a public benchmark derived from real-world platform requests and revised by six music experts. It covers both explicitly stated constraints and expert-validated intent completed from musical references and context.
- We propose **MIRA**, a musical intent refinement agent that combines tool-grounded objective construction, requirement-level audio verification, budgeted prompt search, and feedback memory without modifying the generation backend.
- We validate the proposed evaluation through agreement with expert judgments and demonstrate MIRA across two open-source generators. With commercial systems as references, our results show that MIRA consistently improves intent alignment and enables an open-source generator to match some representative commercial systems.

2 RELATED WORK

Music Generation Evaluation and Benchmarks. Music generation evaluation spans distributional fidelity, text–audio correspondence, perceptual quality, and listener preference (Kilgour et al., 2019; Elizalde et al., 2023; Deshmukh et al., 2024; Liu et al., 2025a; Kim et al., 2025; Zhu & Li, 2026). FAD characterizes distribution-level fidelity, whereas CLAP measures broad semantic correspondence between text and audio. Expert ratings and preference datasets capture complementary aspects of perceived quality. Reward-model benchmarks also assess musicality and instruction fol-

lowing under text, lyrics, and reference-audio conditions (Ma et al., 2026). These complementary perspectives characterize system performance; diagnosing intent alignment additionally requires tracing scores to individual musical requirements.

Audio-language models enable more detailed assessment: Audio Flamingo 3 supports audio reasoning, while Music Flamingo specializes in musical properties such as harmony, structure, and timbre (Goel et al., 2025; Ghosh et al., 2025). AQAScore measures alignment through targeted audio questions, and AnyAudio-Judge decomposes instructions into binary rubric items (Kuan et al., 2026; Li et al., 2026). Related text-to-image benchmarks also evaluate verifiable prompt units and examine agreement with human judgments (Hu et al., 2023; Ghosh et al., 2023; Kamath et al., 2025). Our evaluation follows this item-level approach: MuRA-Bench uses expert-revised rubrics derived from real platform requests, separates explicitly stated requirements from intent completed from references and context, and pairs these criteria with MF-AQA audio-question-answering scores.

Agentic Refinement for Music Generation. Text-conditioned audio and music models provide generation backends (Liu et al., 2023; Agostinelli et al., 2023; Copet et al., 2023; Liu et al., 2024; 2025b; Lei et al., 2025; 2026), with controls over tempo, harmony, instrumentation, vocals, and structure (Melechovsky et al., 2024; Gong et al., 2026; Wang et al., 2026). DITTO further optimizes diffusion noise against differentiable musical objectives without retraining the generator at inference time (Novack et al., 2024). These controls motivate a complementary question: how can feedback on generated audio guide revisions to an open-ended musical request?

Agent research offers mechanisms for such refinement. ReAct supports interaction and external tool use (Yao et al., 2023b); Self-Refine and Reflexion use iterative feedback and experience memory (Madaan et al., 2023; Shinn et al., 2023); and Tree of Thoughts and LATS search over evaluated reasoning or action trajectories (Yao et al., 2023a; Zhou et al., 2024). GEPA derives reflective updates from execution trajectories, and GEMS combines persistent memory with domain-specific skills in multimodal generation (Agrawal et al., 2025; He et al., 2026). These methods make feedback, stored experience, and evaluated alternatives available to a planner; music refinement additionally requires observations grounded in the generated audio. MIRA brings tool-grounded intent completion, requirement-level audio feedback, and branch-specific memory into budgeted prompt search, using unmet requirements to guide repairs across music generation backends without updating their parameters.

3 MURA-BENCH: RUBRIC-LEVEL MUSIC INTENT ALIGNMENT

We formulate text-to-music intent alignment as the satisfaction of a prompt-specific musical rubric. Given a request x , its expert-revised gold rubric is

$$R^*(x) = B^*(x) \cup C^*(x),$$

where $B^*(x)$ contains requirements directly supported by the prompt and $C^*(x)$ contains musical traits completed from references, use cases, and contextual information. Each rubric item describes one property that can be independently judged from the generated audio. This formulation accommodates multiple valid musical realizations without requiring a reference track.

3.1 BENCHMARK CONSTRUCTION

MuRA-Bench contains 100 controlled requests derived from anonymized intent seeds collected on Mureka. Seeds requiring unavailable context, containing sensitive information, or lacking stable musical criteria are excluded. The remaining seeds are normalized while preserving their central intent and realistic expression; raw user logs are not released. The benchmark includes partially specified, reference-based, and hybrid requests with multiple simultaneous constraints.

An instruction-following LLM drafts base and completion requirements and assigns dimensions. Six music experts revise the drafts to preserve explicit constraints, ensure that completed traits are musically plausible and supported by the request or reference context, enforce audible and atomic criteria, and remove or merge redundant or unsupported items. Named artists, songs, and albums are translated into general musical traits rather than direct imitation targets. Items are grouped into seven dimensions: style, instrumentation/vocal, mood, rhythm, harmony/melody, structure/energy,

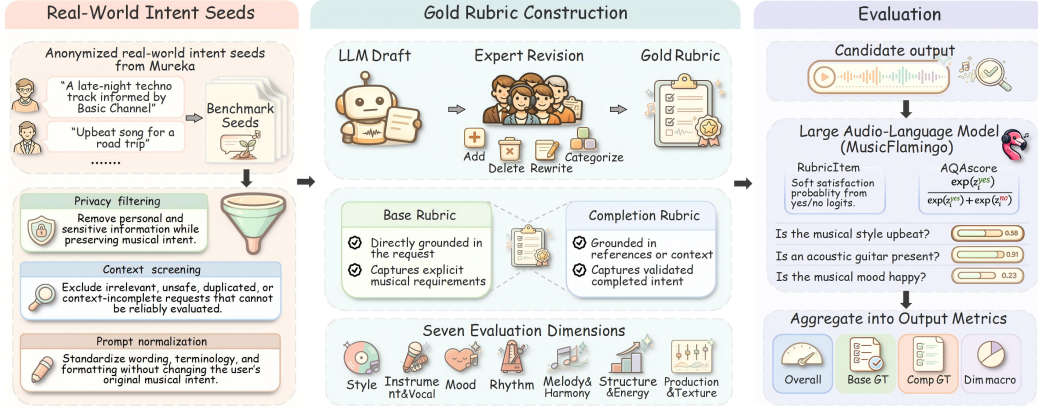


Figure 1: Overview of MuRA-Bench construction and evaluation. Anonymized real-world intent seeds are filtered and transformed into controlled benchmark requests. An LLM drafts base and completion requirements, which are revised by six music experts. During evaluation, MusicFlamingo estimates a soft satisfaction probability for each gold rubric item. Item scores are aggregated into Overall, Dim-Macro, Base GT, and Completion GT. Gold rubrics are used only for evaluation and are not exposed to the generation system.

and production/texture. These categories organize request-specific requirements; they do not impose an identical checklist on every request. Figure 1 summarizes the construction process.

3.2 RUBRIC-BASED EVALUATION

For a generated track a and a gold rubric item r_i , we formulate item-level evaluation as a binary audio question-answering task. The evaluator is instructed to determine whether the audible content satisfies r_i and to answer only *yes* or *no*. Our final evaluator uses MusicFlamingo, a music-specialized audio-language model Ghosh et al. (2025). We compare this design with direct structured judgment and caption-plus-critic alternatives through expert agreement experiments.

Following AQAScore Kuan et al. (2026), let z_i^{yes} and z_i^{no} denote the logits assigned to the two candidate answers. We define the soft satisfaction probability as

$$q_i(a) = \frac{\exp(z_i^{\text{yes}})}{\exp(z_i^{\text{yes}}) + \exp(z_i^{\text{no}})}.$$

The probability retains the evaluator’s confidence and avoids an arbitrary threshold when aggregating item-level judgments.

Let $\mathcal{E}$ denote the set of evaluated request–audio pairs. For a rubric subset $S(x)$, we define its micro-averaged score as

$$\text{Score}(S) = \frac{\sum_{(x,a) \in \mathcal{E}} \sum_{r_i \in S(x)} q_i(a)}{\sum_{(x,a) \in \mathcal{E}} |S(x)|}.$$

Our primary metric is Overall = $\text{Score}(R^*)$. We additionally report Base GT = $\text{Score}(B^*)$ and Completion GT = $\text{Score}(C^*)$. The former measures satisfaction of requirements expressed directly in the benchmark prompt, whereas the latter measures realization of expert-revised traits completed from references and context.

To prevent dimensions containing more rubric items from dominating the analysis, we report **Dim-Macro**. We first micro-average all evaluated items within each musical dimension and then average the resulting seven dimension scores:

$$\text{Dim-Macro} = \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} \frac{1}{|\mathcal{I}_d|} \sum_{(a,r_i) \in \mathcal{I}_d} q_i(a),$$

where $\mathcal{I}_d$ is the set of evaluated items assigned to dimension d . No manual weighting is applied.

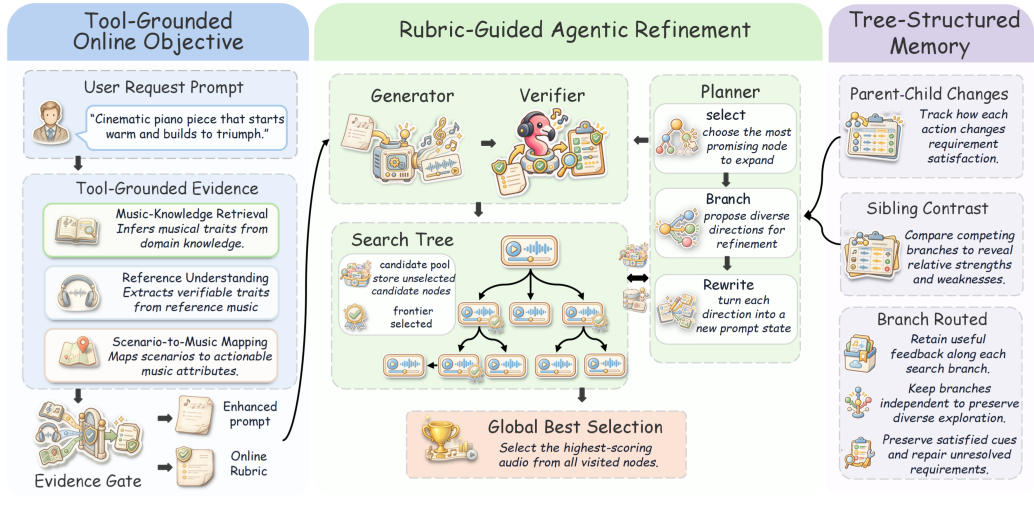


Figure 2: Overview of MIRA and its verifier-guided refinement process.

For grader validation, music experts provide binary item-level labels. An item is marked as satisfied only when sufficient audible evidence is present; uncertain or unsupported cases are conservatively counted as misses. These human labels are used only to measure grader agreement. The expert gold rubrics and human grading labels are never provided to the generation or refinement system.

4 MIRA: MUSICAL INTENT REFINEMENT AGENT

We introduce MIRA (Musical Intent Refinement Agent), a verifier-guided agentic framework that collaborates with any black-box text-to-music generator to improve intent-alignment. As shown in Figure 2, MIRA is built on three core principles: 1) externalizing latent user intent to grounded, objective, and verifiable rubrics; 2) using tree-search to balance exploration and exploitation; 3) maintaining branch-specific memory to preserve rich and the most relevant optimization history.

Starting from a user prompt x and a black-box text-to-music generator G , MIRA first uncovers the implicit intents in the prompt and translates it into a set of grounded rubrics R , together with the musical control parameters κ required by the backend generator. Then, MIRA performs a verifier-guided search process to iteratively improve the generation prompt. Specifically, it constructs a search tree $\mathcal{T} = (\mathcal{V}, \mathcal{E}_{\mathcal{T}})$, where each node $v \in \mathcal{V}$ stores

$$n_v = (p_v, a_v, o_v, M_v),$$

with p_v denoting complete generation prompt, $a_v \sim G(\cdot \mid p_v, \kappa)$ the generated audio sample, o_v the rubric-level judgement produced by a music-specialized verifier, and M_v the branch-specific memory accumulated along the path to v .

4.1 TOOL-GROUNDED ONLINE RUBRIC

A user request may contain both directly stated constraints and musical intent that requires external knowledge to interpret. References, usage scenarios, and functional descriptions can imply musical properties that are not fully expressed in the prompt. MIRA preserves the original request as an immutable intent anchor and, when necessary, invokes external tools for music-knowledge retrieval, reference-audio understanding, or scenario-to-music attribute mapping.

Tool outputs first pass through an evidence gate. Only information with a traceable source, supporting evidence, and sufficient confidence is admitted for intent completion. Named references are converted into generalizable, audibly verifiable musical properties.

Given the request x and admitted external evidence E , a text LLM constructs the online musical rubric

$$R(x, E) = \{r_1, \dots, r_m\},$$

where each rubric item r_i represents an atomic musical requirement that can be assessed independently from the generated audio. The rubric is fixed before search and shared by all candidate nodes. It is derived only from the request and admitted evidence, without access to the expert gold rubric used for offline evaluation.

4.2 RUBRIC-GUIDED SEARCH

Text-to-music generation is stochastic, and the same alignment failure may admit several plausible prompt revisions. A single verification result therefore does not uniquely determine the appropriate repair. Moreover, following a single refinement trajectory may improve one rubric item while degrading others that were previously satisfied. MIRA therefore formulates prompt refinement as rubric-guided search under a finite generation budget.

For a node v , MusicFlamingo evaluates every online rubric item and returns an item-level observation together with an aggregate score:

$$\begin{aligned} q_{v,i} &= P_{\text{MF}}(\text{yes} \mid a_v, r_i), \\ o_v &= (q_{v,1}, \dots, q_{v,m}), \\ s_v &= \frac{1}{m} \sum_{i=1}^m q_{v,i}. \end{aligned}$$

The aggregate score supports candidate comparison, while the item-level scores identify satisfied and unresolved parts of the online rubric. Rubric items with $q_{v,i} < \eta$, where $\eta = 0.60$, are treated as low-scoring items and may receive concise targeted feedback.

At each search layer, the agent uses each retained parent’s prompt, item-level observation, and branch memory to propose several complete candidate prompts. Candidates expanded from the same frontier are generated and verified independently; outcomes are shared only after all candidates at that layer have been evaluated.

After evaluating a layer, MIRA retains a bounded search frontier. It preserves the candidate with the highest aggregate score and a complementary candidate with stronger mean satisfaction over its lowest-scoring quarter of rubric items; any remaining positions are filled according to aggregate score. The observed rubric state thus determines which candidates are retained and which items are targeted by subsequent repairs, while the search topology remains controlled by the generation budget.

Let $\mathcal{V}_B$ denote all nodes generated and evaluated under budget B , including the initial generation. The budget constraint is

$$|\mathcal{V}_B| \leq B.$$

We evaluate a compact setting with $B = 3$ and a broader setting with $B = 9$. Both use the same search mechanism and differ only in the available generation budget.

4.3 TREE-STRUCTURED FEEDBACK MEMORY

Retaining only the highest-scoring candidate does not explain why a revision succeeded or whether it improved some rubric items at the expense of others. MIRA therefore aligns its feedback memory with the search tree, allowing subsequent decisions to use both the local effect of a revision and its performance relative to competing search directions.

For a non-root node v with parent $p(v)$, MIRA first measures the item-level change from its parent:

$$\Delta_{v,i}^{\text{par}} = q_{v,i} - q_{p(v),i}.$$

It then compares the node with the other candidates evaluated at the same search layer:

$$\Delta_{v,i}^{\text{cmp}} = q_{v,i} - \frac{1}{|\mathcal{N}_l(v)|} \sum_{u \in \mathcal{N}_l(v)} q_{u,i},$$

where $\mathcal{N}_l(v)$ denotes the other candidates visible at layer l . The contrastive term is omitted when this set is empty. The parent–child difference captures the local effect of a revision, while the candidate contrast identifies the relative strengths and weaknesses of its search direction.

To reduce sensitivity to minor verifier fluctuations, changes whose absolute magnitude does not exceed $\epsilon = 0.03$ are treated as stable; larger positive and negative changes are recorded as improvements and regressions, respectively. MIRA summarizes these signals as branch-specific memory M_v , recording which rubric items should be preserved, which remain unresolved, and which revision directions have produced regressions. The text LLM uses this memory when proposing subsequent candidate prompts.

Memory is propagated only along the corresponding branch. Different branches do not access one another’s complete prompts or private trajectories, preserving diversity across horizontal search while allowing feedback to persist within each branch. All memory is reset between user requests.

4.4 CANDIDATE SELECTION

Within budget B , MIRA returns the highest-scoring successfully visited candidate:

$$v^* = \arg \max_{v \in \mathcal{V}_B} s_v, \quad a^* = a_{v^*}.$$

Aggregate satisfaction is the primary selection criterion. Ties are resolved using the rubric-item pass rate, lower-tail satisfaction, the minimum item score, and finally generation order.

5 EXPERIMENTS

Implementation Details. We evaluate MIRA on ACE-Step v1.5 Turbo (Gong et al., 2026) and SongGeneration2 Large (Lei et al., 2026), two open-source backends with different generation architectures and initial alignment levels. Kimi 2.6 provides the text planner through its official API, and MusicFlamingo serves as the requirement-level verifier (Ghosh et al., 2025). The generators and verifier run on a single NVIDIA A800. Commercial references use direct prompts through official APIs: MiniMax music-2.6 and music-3.0, Suno v4/v4.5/v5/v5.5, and Mureka V9. We evaluate MIRA with $B = 3$ and $B = 9$, including the initial generation. Main results cover all 100 MuRA-Bench requests; the component analysis uses a fixed, category-stratified subset of 25 requests. The expert-agreement study and system-level human evaluation are reported separately below.

Agreement with Expert Judgments. We assess how well automatic rubric scores agree with expert judgments. Three music evaluators label the same samples without seeing system identities or automatic scores. Evaluation uses the common support of 1,399 rubric items, 99 clips, and 25 requests. CLAPScore provides a global audio–text embedding baseline; Prompting asks MusicFlamingo whether an item is satisfied; Cascade passes MusicFlamingo’s requirement-conditioned audio description to a text judge; and MF-AQA uses normalized yes/no logits for item-level scoring. This comparison tests whether audio question answering (Kuan et al., 2026) supports the music-specific requirements in MuRA-Bench.

Table 1 shows that MF-AQA achieves the highest agreement among the compared evaluators across all six metrics, including item-level SRCC of .690 and pair accuracy of .796, supporting its use for the music-specific requirements in this study.

Main Results. Table 2 first uses MuRA-Bench as a unified diagnostic benchmark to evaluate two open-source backends and seven commercial model versions. The results reveal distinct capability profiles across explicit requirements, completed intent, and the seven musical dimensions. Commercial systems generally exhibit stronger intent alignment, and the Suno family shows a clear progression in Overall performance across successive versions. However, these gains are not uniform across dimensions: the system with the highest Overall score is not consistently the strongest in instrumentation and vocals, harmony and melody, or every other musical category. MuRA-Bench therefore provides not only an aggregate system ranking, but also a fine-grained diagnosis of capability differences that would be obscured by a single global score.

MIRA consistently improves intent alignment on both open-source backends. Increasing the generation budget from $B = 3$ to $B = 9$ further improves Overall and D-Macro, while benefiting both explicitly stated requirements and completed musical intent. Relative to direct generation, MIRA with $B = 9$ yields relative Overall improvements of **12.9%** on ACE-Step and **19.0%** on SongGeneration2. The nearly one-fifth gain on SongGeneration2 further suggests that verifier-guided search

Evaluator	Item-level		Clip-level			Pair
	SRCC	τ_b	LCC	SRCC	τ_b	Acc.
CLAPScore	0.245	0.175	0.361	0.331	0.223	0.626
Prompting	0.552	0.460	0.732	0.743	0.567	0.728
Cascade	0.436	0.350	0.563	0.608	0.434	0.578
MF-AQA (Ours)	0.690	0.532	0.817	0.823	0.646	0.796

Table 1: Agreement between automatic evaluators and expert rubric ratings. Item-level SRCC and τ_b compare evaluator scores with mean expert ratings for individual rubric items. Clip-level LCC, SRCC, and τ_b are computed after averaging all rubric-item scores for each audio. LCC and SRCC denote Pearson and Spearman correlation, respectively; τ_b denotes Kendall’s rank correlation with correction for ties. Pair Acc. measures agreement with experts when selecting the better-aligned audio from candidates generated for the same request. Results use the exact common support of 1,399 rubric items, 99 clips, and 25 requests.

Backend	Setting	Overall	D-Macro	Base	Comp.	Style	I/V	Mood	Rhythm	H/M	S/E	P/T
ACE-Step v1.5 Turbo	Direct prompt	.676	.672	.696	.661	.681	.624	.719	.710	.641	.694	.633
	MIRA(B=3)	.742	.741	.771	.711	.746	.695	.812	.754	.714	.728	.737
	MIRA(B=9)	.763	.756	.791	.734	.754	.691	.830	.772	.728	.746	.770
SongGeneration2 Large	Direct prompt	.605	.592	.629	.583	.554	.511	.650	.689	.490	.641	.607
	MIRA(B=3)	.662	.649	.689	.632	.620	.558	.750	.702	.562	.677	.670
	MIRA(B=9)	.720	.708	.753	.684	.751	.635	.772	.746	.598	.721	.734
MiniMax music-2.6	Direct prompt	.759	.754	.782	.740	.760	.689	.816	.791	.675	.783	.764
MiniMax music-3.0	Direct prompt	.770	.756	.798	.748	.759	.723	.831	.771	.674	.751	.786
Suno v4	Direct prompt	.736	.728	.758	.729	.745	.617	.807	.762	.647	.773	.747
Suno v4.5	Direct prompt	.768	.760	.800	.732	.782	.694	.843	.768	.659	.802	.770
Suno v5	Direct prompt	.797	.790	.831	.760	.817	.725	.853	.810	.735	.806	.786
Suno v5.5	Direct prompt	.819	.809	.853	.777	.848	.753	.869	.833	.709	.846	.804
Mureka V9	Direct prompt	.806	.800	.840	.772	.838	.755	.867	.818	.731	.805	.784

Table 2: Results on the full MuRA-Bench benchmark. D-Macro denotes dimension-macro; Base and Comp. denote satisfaction of explicitly stated and completed intent, respectively. I/V, H/M, S/E, and P/T denote instrumentation/vocal, harmony/melody, structure/energy, and production/texture. Bold marks the best result within each open-source backend or among direct commercial systems.

is particularly effective for backends with weaker initial prompt following. Table 2 shows that these improvements are distributed across multiple musical dimensions rather than being concentrated in a single category.

After refinement, ACE-Step numerically matches MiniMax music-2.6, outperforms Suno v4, and approaches MiniMax music-3.0 and Suno v4.5 in intent alignment. This result indicates that external intent completion, requirement-level music verification, and budgeted search can substantially narrow the alignment gap between open-source generators and commercial systems. More recent Suno versions and Mureka V9 nevertheless remain stronger overall, suggesting that inference-time refinement does not fully eliminate differences in the underlying music generation capability. Because the internal sampling procedures and inference budgets of commercial systems are not controllable, these results compare the alignment levels reached by different systems rather than constitute a strictly budget-matched comparison.

Blinded expert ratings of generated tracks on 25 requests (Figure 3) complement the item-level evaluator validation with a system-level assessment.

Component Analysis. Figure 4 analyzes the contributions of sampling, tool grounding, adaptive search, and tree-structured memory on a fixed subset of 25 requests, proportionally stratified by category from the full MuRA-Bench. Except for direct generation with $B = 1$, all configurations use the same generation budget of $B = 9$. Best-of-9 independently samples nine candidates and selects the one with the highest MusicFlamingo verifier score, providing an equal-budget random-search baseline.

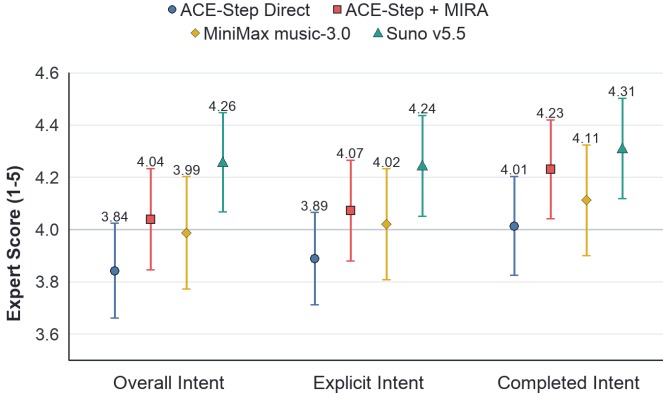


Figure 3: Blinded expert ratings on 25 MuRA-Bench requests. We compare direct ACE-Step, ACE-Step with MIRA, and MiniMax music-3.0 and Suno v5.5. The plot shows overall, explicit, and completed intent ratings on a five-point scale; error bars denote 95% confidence intervals.

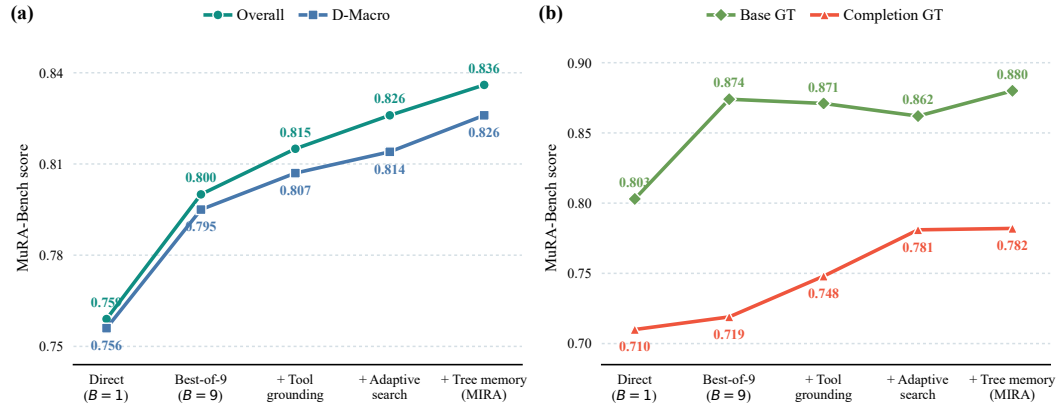


Figure 4: Component analysis on a fixed subset of 25 MuRA-Bench requests, proportionally stratified by category. Direct generation uses $B = 1$, whereas all remaining configurations use $B = 9$. Best-of-9 is an equal-budget random-sampling baseline; tool grounding, verifier-guided adaptive search, and tree-structured memory progressively form the full MIRA framework.

Best-of-9 improves Overall from .759 to .800, but remains substantially below the .836 achieved by full MIRA. The difference is more pronounced on completed intent, where Best-of-9 reaches only .719 compared with MIRA’s .782. Under the same generation budget, MIRA therefore improves over Best-of-9 by 3.6 percentage points in Overall and 6.3 points in Completion, showing that its gains cannot be explained by drawing and selecting more samples alone.

Adding tool grounding to Best-of-9 increases Overall to .815 and Completion to .748, indicating that externally grounded task specifications provide a stronger search target. Replacing independent sampling with verifier-guided adaptive search further raises these scores to .826 and .781, respectively, demonstrating that item-level feedback allocates the same budget more effectively. Finally, tree-structured memory improves Overall from .826 to .836, D-Macro from .814 to .826, and Base from .862 to .880 while preserving the Completion gain. These results identify complementary roles for the three components: tool grounding defines a more complete objective, adaptive search uses rubric-level feedback to explore promising revisions, and tree memory retains branch-specific evidence across subsequent decisions.

6 CONCLUSION

We presented a unified framework for evaluating and improving intent alignment in text-to-music generation. MuRA-Bench converts requests derived from real-world platform usage into expert-revised musical rubrics, enabling generated music to be evaluated at the level of individual requirements. The rubrics cover both explicitly stated constraints and completed musical intent. Our evalu-

ator study further shows that requirement-targeted audio question answering provides a reliable and interpretable approximation of expert rubric judgments.

Building on this formulation, MIRA treats music generation as verifier-guided search in text space. It constructs an online intent specification, evaluates generated audio against individual musical requirements, explores complementary bounded prompt modifications, and uses a music-specialized verifier to select the strongest candidate. Because the entire loop operates outside the generator, MIRA is compatible with structurally different black-box text-to-music backends. Experiments on open-source backends, with commercial systems as references, show that MIRA consistently improves intent alignment and enables an open-source generator to match some representative commercial systems.

REFERENCES

- Andrea Agostinelli, Timo I. Denk, Zalán Borsos, Jesse Engel, Mauro Verzetti, Antoine Caillon, Qingqing Huang, Aren Jansen, Adam Roberts, Marco Tagliasacchi, Matt Sharifi, Neil Zeghidour, and Christian Frank. MusicLM: Generating music from text, 2023.
- Lakshya A. Agrawal, Shangyin Tan, Dilara Soylu, Noah Ziemis, Rishi Khare, Krista Opsahl-Ong, Arnav Singhvi, Herumb Shandilya, Michael J. Ryan, Meng Jiang, Christopher Potts, Koushik Sen, Alexandros G. Dimakis, Ion Stoica, Dan Klein, Matei Zaharia, and Omar Khat-tab. GEPA: Reflective prompt evolution can outperform reinforcement learning. *arXiv preprint arXiv:2507.19457*, 2025. URL <https://arxiv.org/abs/2507.19457>.
- Jade Copet, Felix Kreuk, Itai Gat, Tal Remez, David Kant, Gabriel Synnaeve, Yossi Adi, and Alexandre Défossez. Simple and controllable music generation, 2023.
- Soham Deshmukh, Dareen Alharthi, Benjamin Elizalde, Hannes Gamper, Mahmoud Al Ismail, Rita Singh, Bhiksha Raj, and Huaming Wang. PAM: Prompting audio-language models for audio quality assessment. In *Proceedings of Interspeech*, pp. 3320–3324, 2024. doi: 10.21437/Interspeech.2024-325. URL https://www.isca-archive.org/interspeech_2024/deshmukh24b_interspeech.html.
- Benjamin Elizalde, Soham Deshmukh, Mahmoud Al Ismail, and Huaming Wang. CLAP: Learning audio concepts from natural language supervision. In *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing*, 2023. doi: 10.1109/ICASSP49357.2023.10095889.
- Dhruba Ghosh, Hanna Hajishirzi, and Ludwig Schmidt. GenEval: An object-focused framework for evaluating text-to-image alignment, 2023.
- Sreyan Ghosh, Arushi Goel, Lasha Koroshinadze, Sang-gil Lee, Zhifeng Kong, Joao Felipe Santos, Ramani Duraiswami, Dinesh Manocha, Wei Ping, Mohammad Shoeybi, and Bryan Catanzaro. Music flamingo: Scaling music understanding in audio language models. *arXiv preprint arXiv:2511.10289*, 2025. URL <https://arxiv.org/abs/2511.10289>.
- Arushi Goel, Sreyan Ghosh, Jaehyeon Kim, Sonal Kumar, Zhifeng Kong, Sang-gil Lee, Chao-Han Huck Yang, Ramani Duraiswami, Dinesh Manocha, Rafael Valle, and Bryan Catanzaro. Audio flamingo 3: Advancing audio intelligence with fully open large audio language models. *arXiv preprint arXiv:2507.08128*, 2025. URL <https://arxiv.org/abs/2507.08128>.
- Junmin Gong, Yulin Song, Wenxiao Zhao, Sen Wang, Shengyuan Xu, and Jing Guo. ACE-Step 1.5: Pushing the boundaries of open-source music generation, 2026.
- Zefeng He, Siyuan Huang, Xiaoye Qu, Yafu Li, Tong Zhu, Yu Cheng, and Yang Yang. GEMS: Agent-native multimodal generation with memory and skills. *arXiv preprint arXiv:2603.28088*, 2026. URL <https://arxiv.org/abs/2603.28088>.
- Dorien Herremans and Abhinaba Roy. Aligning generative music ai with human preferences: Methods and challenges. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pp. 39699–39706, 2026. doi: 10.1609/aaai.v40i46.41323.

- Yushi Hu, Benlin Liu, Jungo Kasai, Yizhong Wang, Mari Ostendorf, Ranjay Krishna, and Noah A. Smith. TIFA: Accurate and interpretable text-to-image faithfulness evaluation with question answering. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023.
- Amita Kamath, Kai-Wei Chang, Ranjay Krishna, Luke Zettlemoyer, Yushi Hu, and Marjan Ghazvininejad. GenEval 2: Addressing benchmark drift in text-to-image evaluation, 2025.
- Kevin Kilgour, Mauricio Zuluaga, Dominik Roblek, and Matthew Sharifi. Fréchet audio distance: A reference-free metric for evaluating music enhancement algorithms. In *Proceedings of Interspeech*, 2019.
- Yonghyun Kim, Wayne Chi, Anastasios Nikolas Angelopoulos, Wei-Lin Chiang, Koichi Saito, Shinji Watanabe, Yuki Mitsufuji, and Chris Donahue. Music arena: Live evaluation for text-to-music. In *NeurIPS Creative AI Track*, 2025.
- Junghyun Koo, Gordon Wichern, François G. Germain, Sameer Khurana, and Jonathan Le Roux. SMITIN: Self-monitored inference-time intervention for generative music transformers. *IEEE Open Journal of Signal Processing*, 2025. doi: 10.1109/OJSP.2025.3534686.
- Chun-Yi Kuan, Kai-Wei Chang, and Hung-yi Lee. AQAScore: Evaluating semantic alignment in text-to-audio generation via audio question answering, 2026.
- Shun Lei, Yixuan Zhou, Boshi Tang, Max W. Y. Lam, Feng Liu, Hangyu Liu, Jingcheng Wu, Shiyin Kang, Zhiyong Wu, and Helen Meng. SongCreator: Lyrics-based universal song generation. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi: 10.52202/079017-2546.
- Shun Lei, Yaoxun Xu, Zhiwei Lin, Huaicheng Zhang, Wei Tan, Hangting Chen, Jianwei Yu, Yixuan Zhang, Chenyu Yang, Haina Zhu, Shuai Wang, Zhiyong Wu, and Dong Yu. LeVo: High-quality song generation with multi-preference alignment. In *Advances in Neural Information Processing Systems*, volume 38, 2025. URL <https://nips.cc/virtual/2025/poster/118257>.
- Shun Lei, Huaicheng Zhang, Dapeng Wu, Yaoxun Xu, Lishi Zuo, Wei Tan, Hangting Chen, Guangzheng Li, Jianwei Yu, Zhiyong Wu, and Dong Yu. LeVo 2: Stable and melodious song generation via hierarchical representation modeling and progressive post-training. *arXiv preprint arXiv:2606.30642*, 2026. URL <https://arxiv.org/abs/2606.30642>.
- Haitao Li, Tian Tan, Yuguang Yang, Shan Yang, and Xie Chen. AnyAudio-Judge: A dynamic rubric-based benchmark and evaluator for audio instruction following, 2026.
- Cheng Liu, Hui Wang, Jinghua Zhao, Shiwan Zhao, Hui Bu, Xin Xu, Jiaming Zhou, Haoqin Sun, and Yong Qin. MusicEval: A generative music corpus with expert ratings for automatic text-to-music evaluation, 2025a.
- Haohe Liu, Zehua Chen, Yi Yuan, Xinhao Mei, Xubo Liu, Danilo Mandic, Wenwu Wang, and Mark D. Plumbley. AudioLDM: Text-to-audio generation with latent diffusion models. In *Proceedings of the 40th International Conference on Machine Learning*, pp. 21450–21474, 2023. URL <https://proceedings.mlr.press/v202/liu23f.html>.
- Haohe Liu, Yi Yuan, Xubo Liu, Xinhao Mei, Qiuqiang Kong, Qiao Tian, Yuping Wang, Wenwu Wang, Yuxuan Wang, and Mark D. Plumbley. AudioLDM 2: Learning holistic audio generation with self-supervised pretraining. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 2024. doi: 10.1109/TASLP.2024.3399607.
- Zihan Liu, Shuangrui Ding, Zhixiong Zhang, Xiaoyi Dong, Pan Zhang, Yuhang Zang, Yuhang Cao, Dahua Lin, and Jiaqi Wang. SongGen: A single stage auto-regressive transformer for text-to-song generation, 2025b.
- Yinghao Ma, Haiwen Xia, Hwei Gao, Weixiong Chen, Yuxin Ye, Yuchen Yang, Sungkyun Chang, Mingshuo Ding, Yizhi Li, Ruibin Yuan, Simon Dixon, and Emmanouil Benetos. CMI-RewardBench: Evaluating music reward models with compositional multimodal instruction, 2026.

- Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegrefe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder, Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, and Peter Clark. Self-refine: Iterative refinement with self-feedback. In *Advances in Neural Information Processing Systems*, 2023.
- Jan Melechovsky, Zixun Guo, Deepanway Ghosal, Navonil Majumder, Dorien Herremans, and Soujanya Poria. Mustango: Toward controllable text-to-music generation. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pp. 8293–8316, Mexico City, Mexico, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.naacl-long.459.
- Zachary Novack, Julian McAuley, Taylor Berg-Kirkpatrick, and Nicholas J. Bryan. DITTO: Diffusion inference-time t-optimization for music generation. In *Proceedings of the International Conference on Machine Learning*, 2024.
- Abhinaba Roy, Geeta Puri, and Dorien Herremans. Text2midi-InferAlign: Improving symbolic music generation with inference-time alignment, 2025.
- Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 36, 2023.
- Yuejiao Wang, Zihao Ji, Pengfei Cai, Xu Li, Haorui Zheng, Zewen Song, Zhongliang Liu, Chen Zhang, and Pengfei Wan. SegTune: Structured and fine-grained control for song generation, 2026.
- Dapeng Wu, Shun Lei, Wei Tan, Guangzheng Li, Yunzhe Wang, Huaicheng Zhang, Lishi Zuo, and Zhiyong Wu. SongBench: A fine-grained multi-aspect benchmark for song quality assessment, 2026.
- Chenyu Yang, Shuai Wang, Hangting Chen, Wei Tan, Jianwei Yu, and Haizhou Li. SongBloom: Coherent song generation via interleaved autoregressive sketching and diffusion refinement. In *Advances in Neural Information Processing Systems*, volume 38, 2025.
- Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas L. Griffiths, Yuan Cao, and Karthik Narasimhan. Tree of thoughts: Deliberate problem solving with large language models. In *Advances in Neural Information Processing Systems*, volume 36, pp. 11809–11822, 2023a. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/271db9922b8d1f4dd7aaef84ed5ac703-Abstract-Conference.html.
- Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing reasoning and acting in language models. In *Proceedings of the International Conference on Learning Representations*, 2023b.
- Andy Zhou, Kai Yan, Michal Shlapentokh-Rothman, Haohan Wang, and Yu-Xiong Wang. Language agent tree search unifies reasoning, acting, and planning in language models. In *Proceedings of the 41st International Conference on Machine Learning*, pp. 62138–62160, 2024. URL <https://proceedings.mlr.press/v235/zhou24r.html>.
- Di Zhu and Zixuan Li. MuQ-Eval: An open-source per-sample quality metric for ai music generation evaluation, 2026.

A CASE STUDY ILLUSTRATION

Figure 5 presents a complete agent trace for one request. The trace exposes the observation–action–selection loop: the verifier first identifies which rubric items are satisfied or missing in the initial generation; MIRA then converts different failure hypotheses into two branch actions, a conservative repair that preserves the stable techno pulse and bass pressure while adding the missing dub delay, and an exploratory repair that reformulates the request as a more minimal dub-techno texture. Both children are evaluated against the same online rubric, and the final output is selected by verifier support rather than by the language model’s self-preference. This illustrates that MIRA performs targeted branching over repair actions, not only repeated sampling from a single expanded prompt.

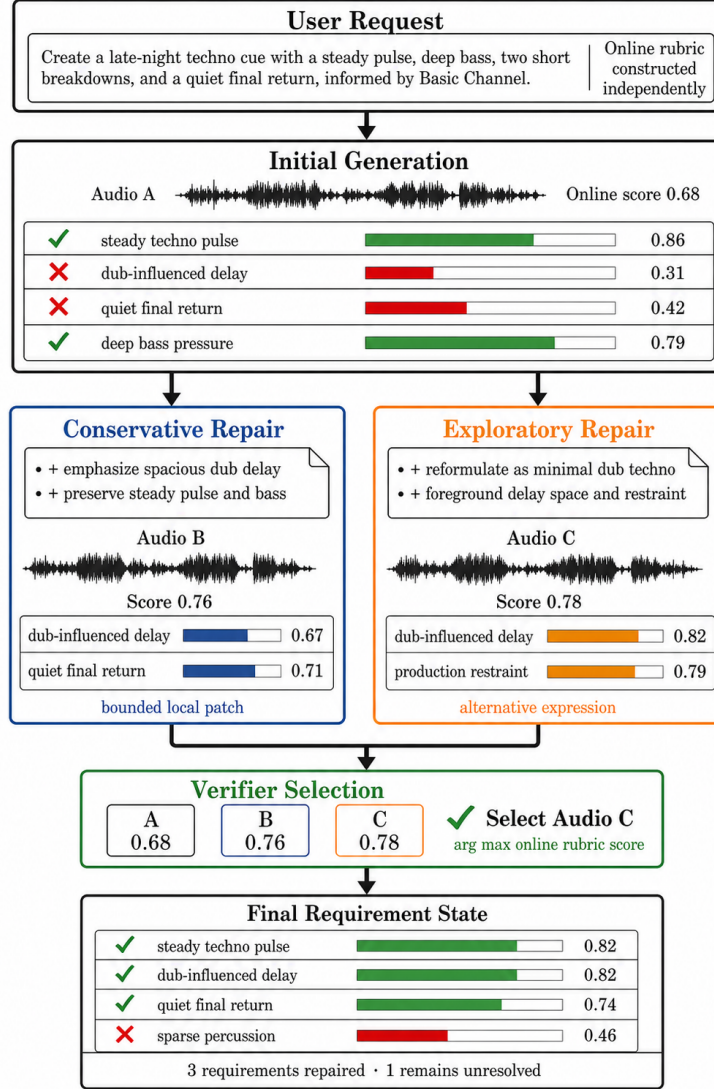


Figure 5: Case study of MIRA’s diversified refinement. The online verifier identifies low-scoring requirements in the initial generation, after which MIRA produces conservative and exploratory prompt repairs. The same verifier scores all candidates and selects the strongest-supported output. Item values are MusicFlamingo satisfaction probabilities.